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Correct. The feminist scholars mentioned in the first paragraph are concerned with women of different social classes and religions, whereas Krontiris's work focuses on a limited social group.

The correct answer is E.

•

According to the passage, feminist scholars cite Burckhardt's view of Renaissance women primarily for which of the following reasons?

A. Burckhardt's view forms the basis for most arguments refuting Kelly's point of view.

B. Burckhardt's view has been discredited by Kelly.

C. Burckhardt's view is one that many feminist scholars wish to refute.

D. Burckhardt's work provides rich historical evidence of inequality between Renaissance women and men.

E. Burckhardt's work includes historical research supporting the arguments of the feminist scholars.

Supporting Idea

This question asks what the passage says explicitly about why feminist scholars reference Burckhardt's view of Renaissance women. The first paragraph states that Burckhardt's view is that Renaissance women enjoyed *perfect equality* with men, and then follows that by noting how feminist scholars have *repeatedly cited* this view to contrast it with extensive evidence of women's inequality during the Renaissance.

A. The passage does not indicate that any feminist scholars cite Burckhardt to refute Kelly's view. It uses Krontiris as an example of scholars who refute Kelly's point of view to a certain degree, but Krontiris does not use Burckhardt's view as her basis for doing so; Krontiris argues against Burckhardt as well.

B. According to the first paragraph, Kelly's work was in certain ways inconsistent with Burckhardt's view, but that is not a reason why Burckhardt's view is cited by feminist scholars. Rather, according to the passage, they cite it in order to argue against it.

C. **Correct.** Many feminist scholars wish to refute Burckhardt's view that Renaissance women and men were equal.

D. As the first paragraph makes clear, Burckhardt's work emphasizes equality, not inequality, between Renaissance women and men.

E. The passage does not discuss the historical research on which Burckhardt based his work.

The correct answer is C.

•

It can be inferred that both Burckhardt and Kelly have been criticized by the scholars mentioned in line 12 for which of the following?

A. Assuming that women writers of the Renaissance are representative of Renaissance women in general

B. Drawing conclusions that are based on the study of an atypical group of women

C. Failing to describe clearly the relationship between social status and literacy among Renaissance women

D. Failing to acknowledge the role played by Renaissance women in opposing cultural stereotypes

E. Failing to acknowledge the ways in which social status affected the creative activities of Renaissance women

Inference

Line 12 refers to feminist scholars who have rejected both Kelly's and Burckhardt's views of the status of Renaissance women. The next sentence states that the feminist scholars use class and religious differences among Renaissance women to argue against Kelly's and Burckhardt's generalizations, which were based on upper-class Italian women.

A. The second paragraph suggests that Krontiris at times conflates Renaissance women writers and women in general, but the passage does not indicate that the feminist scholars believe this of Kelly or Burckhardt.

B. **Correct.** The feminist scholars mentioned study different types of Renaissance women and so reject Kelly's and Burckhardt's conclusions that were based on a group that was not in fact typical.

C. Krontiris, not Kelly and Burckhardt, is the scholar who, according to the passage, fails to address the

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C. Krontiris, not Kelly and Burckhardt, is the scholar who, according to the passage, fails to address the relationship between literacy and social status.

D. The passage provides no grounds for determining whether Kelly, Burckhardt, or the feminist scholars mentioned in the first paragraph dealt with Renaissance women's opposition to cultural stereotypes; Krontiris's work is concerned with this question.

E. The first paragraph suggests that feminist scholars criticized Kelly and Burckhardt for failing to acknowledge the ways in which social status complicates any generalizations that can be made about Renaissance women's lives, not their creative activities specifically.

The correct answer is B.

•

The author of the passage suggests that Krontiris incorrectly assumes that

A. social differences among Renaissance women are less important than the fact that they were women

B. literacy among Renaissance women was more prevalent than most scholars today acknowledge

C. during the Renaissance, women were able to successfully oppose cultural stereotypes relating to gender

D. Renaissance women did not face many difficult social obstacles relating to their gender

E. in order to attain power, Renaissance women attacked basic assumptions in the ideologies that oppressed them

Inference

The first statement the author makes about Krontiris, in the second paragraph, concerns what the author characterizes as a problem with Krontiris's work. Krontiris takes the Renaissance women writers she studies as representative of all Renaissance women; the author says that designating *women* as the most important grouping fails to consider whether other social differences might make for differences in experience.

A. **Correct.** The author indicates that Krontiris's error lies in assuming that women's identity as women trumps social and other differences.

B. The author does not suggest that Krontiris assumes inappropriate literacy levels among Renaissance women, but rather that Krontiris does not give sufficient consideration to the idea that women who could read and write most likely led lives very different from those of women who could not read and write.

C. The author says that Krontiris suggests that there were many cultural stereotypes that women were not able to oppose effectively.

D. Krontiris, according to the author, acknowledges the many social obstacles faced by women on the basis of their gender.

E. According to the author, Krontiris's concluding remarks suggest that Renaissance women *seldom attacked the basic assumptions in the ideologies that oppressed them.*

The correct answer is A.

•

The last sentence in the passage serves primarily to

A. suggest that Krontiris's work is not representative of recent trends among feminist scholars

B. undermine the argument that literate women of the Renaissance sought to oppose social constraints imposed on them

C. show a way in which Krontiris's work illustrates a "cautiously optimistic" assessment of Renaissance women's achievements

D. summarize Krontiris's view of the effect of literacy on the lives of upper- and middle-class Renaissance women

E. illustrate the way in which Krontiris's study differs from the studies done by Burckhardt and Kelly

Evaluation

The function of the final sentence of the passage is to indicate how Krontiris's work takes neither a completely positive nor completely negative view of Renaissance women's experiences—i.e., how her work is representative of those authors who are cautiously optimistic about the achievements of Renaissance women.

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A. The passage discusses Krontiris's work as an example of the trend described in the latter part of the first paragraph and mentioned in the first line of the second paragraph. The last sentence in the passage shows that Krontiris's work is in fact representative of recent trends among feminist scholars.

B. The last sentence in the passage states that Renaissance women's opposition to cultural stereotypes was circumscribed, but it also suggests that these women did gain some power for themselves. Thus, the sentence does not serve primarily to undermine the argument that the women sought to oppose social constraints.

C. **Correct.** Krontiris's work illustrates the "cautiously optimistic" view by embracing both the idea that Renaissance women could gain a certain amount of power and the idea that the extent of their opposition was limited.

D. The last sentence in the passage summarizes Krontiris's view, but that view does not, according to the passage, take into account the effect of literacy on the members of a particular social class.

E. The main function of the final sentence of the passage is to take up the idea of the "cautiously optimistic" assessment offered in the penultimate sentence. This does mark a significant departure from both Burckhardt and Kelly, but the distinction between their work and that of other feminist scholars is marked more clearly earlier in the passage.

The correct answer is C.

•

The author of the passage implies that the women studied by Krontiris are unusual in which of the following ways?

A. They faced obstacles less formidable than those faced by other Renaissance women.

B. They have been seen by historians as more interesting than other Renaissance women.

C. They were more concerned about recording history accurately than were other Renaissance women.

D. Their perceptions are more likely to be accessible to historians than are those of most other Renaissance women.

E. Their concerns are likely to be of greater interest to feminist scholars than are the ideas of most other Renaissance women.

Inference

The women Krontiris studied are unusual, the author suggests, because they were literate, thus putting them among the minority of Renaissance women. That they could write, however, means that their written reflections are part of the historical record, whereas the direct impressions of experiences had by Renaissance women who could not write about their lives are lost to history.

A. The author implies that the obstacles faced by Krontiris's subjects may have been different from those faced by other women, not that they were less formidable.

B. The author does not imply that the women studied by Krontiris are seen as more interesting; rather, the author indicates that their work is that which is available for study.

C. The women Krontiris studies were able to record their own history because they, unlike most other Renaissance women, were literate. This does not imply that they were more concerned with recording history accurately.

D. **Correct.** Because Krontiris's subjects were literate, they were able to write down, and thus preserve for historians, their perceptions in a way that most other Renaissance women were not.

E. The author does not suggest that feminist scholars in general are more interested in the concerns of middle- and upper-class literate women than they are with women of other classes.

The correct answer is D.

Questions 606–609 refer to the passage on page 497.

Main Idea Summary

The main idea of the passage is that observations of asteroid collisions and computer simulations of such collisions indicate that most observed asteroids are loose collections of rocks and only small asteroids are individual, solid rocks. The passage begins by describing how, in theory, one could determine the structure of asteroids by observing the rates at which they spin after collisions. The passage then reports that, using this method, scientists have found that most observed asteroids spin in ways that are inconsistent with their being solid rocks. The second paragraph indicates that the conclusions are supported by computer modeling and then presents a causal explanation for the

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observed phenomena.

606. The passage implies which of the following about the five asteroids mentioned in line 12?

A. Their rotation rates are approximately the same.

B. They have undergone approximately the same number of collisions.

C. They are monoliths.

D. They are composed of fragments that have escaped the gravity of larger asteroids.

E. They were detected only recently.

Inference

In line 12, *five observed asteroids* refers to the five asteroids whose rotation rates are exceptions to the strict limit on the rate of rotation found in all other observed asteroids. These five asteroids all have diameters smaller than 200 meters. The passage indicates that if asteroids were all monoliths—that is, single rocks—then their rotation rates would form a bell curve when graphed, but if asteroids were piles of rubble, the tail of the bell curve indicating very fast rotation rates would be missing. Among asteroids larger than 200 meters, this tail is missing, and only the five asteroids described as exceptions have rotation rates falling at the very high end of the bell curve.

A. All that the passage states about the rotation rates of these five asteroids is that they do not obey a strict limit. The passage does not rule out that their rates of rotation are significantly different from one another.

B. According to the passage, frequent collisions occur among asteroids. But the passage does not suggest that asteroids that are of similar sizes, or that have particularly high rotation rates, will be similar in terms of the number of collisions that they have undergone to reach those distinctive states.

C. **Correct.** The second paragraph states that *most small asteroids* should be monolithic, and the five observed asteroids are all smaller than 200 meters in diameter.

D. The five asteroids are most likely not composed of fragments because, as the passage states, small asteroids should be monoliths.

E. The passage notes that researchers have observed these five asteroids, along with others, but it does not indicate when these asteroids were originally detected.

The correct answer is C.

607. The discovery of which of the following would call into question the conclusion mentioned in line 16?

A. An asteroid 100 meters in diameter rotating at a rate of once per week

B. An asteroid 150 meters in diameter rotating at a rate of 20 times per hour

C. An asteroid 250 meters in diameter rotating at a rate of once per week

D. An asteroid 500 meters in diameter rotating at a rate of once per hour

E. An asteroid 1,000 meters in diameter rotating at a rate of once every 24 hours

Application

The conclusion that the text in line 16 points to is that asteroids with diameters greater than 200 meters are *multicomponent structures or rubble piles*. To call that conclusion into question, an observation would have to suggest that asteroids larger than 200 meters across are not such multicomponent structures. According to the first paragraph, rubble piles cannot be fast rotators: spinning faster than once every few hours would make them fly apart.

A. Nothing in the passage suggests that the behavior of an asteroid 100 meters in diameter is relevant to a conclusion about the behavior of asteroids greater than 200 meters in diameter.

B. Nothing in the passage suggests that the behavior of an asteroid 150 meters in diameter would have any effect on a conclusion about the constitution of asteroids with diameters greater than 200 meters.

C. An asteroid 250 meters in diameter rotating at a rate of once per week would be rotating at a slow enough rate to hold together a pile of rubble. Thus, this observation would be entirely consistent with the conclusion about asteroids larger than 200 meters in diameter.

D. **Correct.** Assuming that an asteroid composed of a pile of rubble is of a great enough density, a rotation rate

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D. **Correct.** Assuming that an asteroid composed of a pile of rubble is of a great enough density, a rotation rate greater than one revolution every few hours would make it fly apart. So a 500-meter asteroid rotating at a rate of once per hour—that is, faster than the crucial speed—would fly apart if it were not a monolith. The conclusion states that all asteroids larger than 200 meters are multicomponent structures—that is, are not monoliths—so the discovery of a 500-meter asteroid rotating at a rate of once an hour would call into question that conclusion.

E. An asteroid rotating at a rate of once every 24 hours would, regardless of size, be rotating much more slowly than the *once every few hours* that the passage claims would make a pile of rubble of a sufficient density fly apart. So an asteroid with a diameter of 1,000 meters that rotated once per day could be a pile of rubble and not conflict with the conclusion.

The correct answer is D.

608. According to the passage, which of the following is a prediction that is based on the strength of the gravitational attraction of small asteroids?

A. Small asteroids will be few in number.

B. Small asteroids will be monoliths.

C. Small asteroids will collide with other asteroids very rarely.

D. Most small asteroids will have very fast rotation rates.

E. Almost no small asteroids will have very slow rotation rates.

Supporting Idea

Regarding small asteroids, the second paragraph states that they have feeble gravity. Any fragments from impacts would escape that gravity, and thus, the passage states, the small asteroids *should be monolithic*.

A. Small asteroids could be few in number, but the passage does not offer such a prediction.

B. **Correct.** This prediction is offered in the second paragraph, based on the fact that small asteroids do not have strong gravitational attraction. Any impact fragments will easily escape the weak gravitational attraction of the small asteroids.

C. The passage discusses large asteroids collisions in more detail than small-asteroid collisions, but it provides no basis for predicting how often large and small asteroids will, comparatively, be involved in such collisions.

D. The first paragraph indicates that the rotation rates of small asteroids can exceed the upper limit on the rotation rates of large asteroids, but it does not indicate that most small asteroids have rotation rates that exceed this upper limit.

E. The passage only indicates that there are few observed exceptions to the upper limit on rotation rates of large asteroids, and these exceptions are all smaller than 200 meters in diameter; the passage does not indicate that there are few small asteroids that have very slow rotation rates.

The correct answer is B.

609. The author of the passage mentions “escape velocity” (see line 22) in order to help explain which of the following?

A. The tendency for asteroids to become smaller rather than larger over time

B. The speed with which impact fragments reassemble when they do not escape an asteroid’s gravitational attraction after a collision

C. The frequency with which collisions among asteroids occur

D. The rotation rates of asteroids smaller than 200 meters in diameter

E. The tendency for large asteroids to persist after collisions

Evaluation

This question asks about the purpose of the author’s use of the phrase *escape velocity* in the second paragraph. The author is discussing what occurs after an asteroid collision, in which a large asteroid might be blasted to bits. The bits, according to the author, will move slower than their *mutual escape velocity*—that is, the speed at which they would have to move to get away from each other and not reassemble, under the influence of gravity, into a rubble pile.

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into a rubble pile.

A. The author is emphasizing the asteroid bits that do not escape rather than those that do. Asteroids may become smaller over time, but the fact that most bits move slower than their escape velocity would not help to explain this shrinkage.

B. That the bits of asteroid move slower than their escape velocity helps explain why the fragments reassemble, but it does not help explain the speed with which they reassemble.

C. According to the author, asteroid collisions occur frequently, but the escape velocity of the resulting fragments does not help to explain that frequency.

D. The concept of escape velocity may help explain why small asteroids are monoliths, but it has no relevance, at least as far as the passage indicates, to those asteroids' rotation rates.

E. **Correct.** After a collision, it is the asteroid fragments' failure to reach escape velocity that allows the fragments' gravitational pull to reassemble them into a rubble pile.

The correct answer is E.

Questions 610–612 refer to the passage on page 499.

Main Idea Summary

The primary purpose of the passage is to discuss the theory underlying attempts to get particles to travel faster than the speed of light by means of quantum tunneling. The passage states that quantum theory implies that there is a small probability that a particle can tunnel through a solid barrier. The time it takes a particle to tunnel through a barrier increases as the thickness of the barrier increases, but only up to a maximum amount of time; beyond that maximum, the amount of time does not vary no matter the thickness of the barrier. This implies that, given a certain thickness, the particle will exceed the speed of light, and this has received some empirical support.

610. The author of the passage mentions calculations about tunneling time and barrier thickness in order to

A. suggest that tunneling time is unrelated to barrier thickness

B. explain the evidence by which Wigner and Eisenbud discovered the phenomenon of tunneling

C. describe data recently challenged by Raymond Chiao and colleagues

D. question why particles engaged in quantum tunneling rarely achieve extremely high speeds

E. explain the basis for Wigner and Eisenbud's hypothesis

Evaluation

This question asks why the author discusses calculations about tunneling time and barrier thickness. According to the passage, these calculations provided the grounds for Wigner and Eisenbud's hypothesis that tunneling particles may travel faster than light.

A. The passage states that tunneling time is related to barrier thickness, up to the point at which tunneling time reaches a maximum.

B. The passage indicates that the phenomenon of tunneling was noted at least as early as 1932. It provides no evidence that Wigner and Eisenbud discovered it.

C. The passage uses Chiao's work to support the idea that tunneling particles may move faster than light, not challenge it.

D. The author describes calculations about tunneling time and barrier thickness in order to explain that particles engaged in quantum tunneling may in fact achieve extremely high speeds, not to explain the rarity of the phenomenon.

E. **Correct.** The calculations about tunneling time and barrier thickness supported Wigner and Eisenbud's hypothesis that quantum tunneling could occur at speeds faster than that of light.

The correct answer is E.

611. The passage implies that if tunneling time reached no maximum in increasing with barrier thickness, then

A. tunneling speed would increase with barrier thickness

B. tunneling speed would decline with barrier thickness

C. tunneling speed would vary with barrier thickness

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C. tunneling speed would vary with barrier thickness

D. tunneling speed would not be expected to increase without limit

E. successful tunneling would occur even less frequently than it does

Inference

The passage states that because tunneling time reaches a maximum, then tunneling speed must increase as barrier thickness increases. But if tunneling time did not reach such a maximum, then speed need not increase without limit; the particle could have as low a speed in thicker barriers as in thinner ones and take longer to tunnel through a barrier.

A. If tunneling time could not reach a maximum, then speed might increase, decrease, or remain the same as barrier thickness increases.

B. If tunneling time could not reach a maximum, then speed might increase, decrease, or remain the same as barrier thickness increases.

C. Tunneling speed could vary with barrier thickness if tunneling time could not reach a maximum, but there is no basis in the passage on which to conclude that this is definitely so.

D. **Correct.** The tunneling particle could have as low a speed in thicker barriers as in thinner ones and simply take longer to make its way through a thicker barrier.

E. The passage states that the probability of successful tunneling declines as the thickness of the barrier increases. However, it does not address the issue of whether the differences in probability of successful tunneling are due to the greater time required to go through thicker barriers.

The correct answer is D.

612. Which of the following statements about the earliest scientific investigators of quantum tunneling can be inferred from the passage?

A. They found it difficult to increase barrier thickness continually.

B. They anticipated the later results of Chiao and his colleagues.

C. They did not suppose that tunneling particles could travel faster than light.

D. They were unable to observe instances of successful tunneling.

E. They made use of photons to study the phenomenon of tunneling.

Inference

This question asks about the earliest investigators of quantum tunneling. The passage notes that quantum tunneling's *extreme rapidity* was observed in 1932; thus, the earliest investigators of this phenomenon knew of its existence at that time. Not until 1955 did Wigner and Eisenbud hypothesize that the particles traveled faster than light. Thus, it is logical to infer that the earliest investigators did not imagine such a speed.

A. There is nothing in the passage to suggest that the earliest investigators of quantum tunneling had difficulty manipulating barrier thickness.

B. The passage states that Chiao and his colleagues measured photons moving at 1.7 times the speed of light—but the passage does not provide evidence that the earliest investigators anticipated such speeds.

C. **Correct.** The passage suggests that prior to 1955, investigators of quantum tunneling had not hypothesized that the particles could travel faster than the speed of light.

D. The passage indicates that by 1932, investigators had noted the rapidity of quantum tunneling; although this does not entail that they observed the phenomenon, it is consistent with their having been able to do so.

E. The passage indicates that Chiao's work involves photons, but it does not indicate the type of particles used or observed by the earliest investigators of the phenomenon.

The correct answer is C.

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